

Taha Bekar

Robotics Software Engineer — Perception — SLAM — Autonomous Systems
mtahabekar@gmail.com — +1 (838) 262 0019 — LinkedIn — Github — Portfolio

Education

Rensselaer Polytechnic Institute M.S. Electrical, Computer, and Systems Engineering 2024 – May 2026 Focus: Autonomy, Robotics, and Control	Troy, NY	Middle East Technical University B.S. Electrical and Electronics Engineering 2018 – 2024 GPA: 3.56 / 4.00	Ankara, Turkey
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Research Experience

Deformable Object Manipulation Research – RPI

2025 – Present

Research Assistant / Master's Project

- Developed a Python-based, ROS2-compatible deformable-object simulation framework in NVIDIA Isaac Sim using Newton Style3D for real-time cloth manipulation with anisotropic material, contact, friction, damping, and bending parameters.
- Built an obstacle-aware cloth manipulation pipeline in Python with kinematic corner pinning, simulated cloth state publishing, ROS2 command interfaces, obstacle-distance computation, and closed-loop follower-corner control.
- Integrated a Control Barrier Function safety controller in Python for cloth-obstacle avoidance in contact-rich manipulation scenarios.
- Developed a real-time ROS2 perception and state estimation pipeline in Python for planar deformable objects using RGB-D data from an end-effector-mounted Azure Kinect, ArUco marker tracking, and Kalman filtering; deployed on a Kinova JACO2 arm and Clearpath Dingo mobile base for real-to-sim calibration.
- Developing a centralized PPO-based reinforcement learning framework in NVIDIA Isaac Lab for coordinated dual-arm deformable-object transport, with continuous Cartesian control and privileged robot/object state observations.

Work Experience

Kuartis

Ankara, Turkey

Computer Vision / Machine Learning Engineer

Feb 2024 – Jun 2024

- Developed and benchmarked real-time, production-style C++ implementations of LiDAR point cloud registration algorithms, including ICP, NDT, GICP, and FAST-GICP, for SLAM and mapping systems.
- Implemented real-time LiDAR-IMU sensor fusion pipelines in C++ by combining LiDAR odometry with IMU odometry and evaluating tightly coupled and loosely coupled fusion strategies for improved localization robustness.
- Evaluated C++ implementations of SLAM frameworks including LOAM, LIO-SAM, and FAST-LIO for localization accuracy, runtime performance, and computational efficiency.
- Improved LIO-SAM localization accuracy by 6% by integrating GICP-based scan matching in C++ into the GTSAM factor-graph optimization framework.

GT-ARC Gemeinnützige GmbH, DAI Labor, TU Berlin

Berlin, Germany

Software Engineer Intern

Jul 2023 – Sep 2023

- Developed a production-oriented C++ LiDAR-based free parking space detection algorithm for autonomous vehicles using ROS, Autoware, and HD Maps.
- Implemented geometric intersection analysis in C++ between detected vehicles and mapped parking regions, achieving 86% free space detection accuracy.

Analog Devices

Istanbul, Turkey

Software Engineer Intern

Jul 2022 – Aug 2022

- Developed production-level C firmware for a PWM controller on the MAX32666 microcontroller using hardware timer peripherals; the implementation was incorporated into the official microcontroller firmware.
- Built a GUI interface for validating PWM signals and implemented automated unit tests using Google Test for the production firmware.

Projects

Autonomous UGV Mapping System

- Converted a 2D LiDAR into a 3D scanning system via servo rotation to enable volumetric mapping.
- Implemented OctoMap mapping using ICP (Iterative Closest Point) registration and AMCL (Adaptive Monte Carlo Localization) algorithms in ROS. Achieved accuracy levels of 95% and 75% for mapping and localization, respectively.

Lane Detection System

- Implemented classical vision-based lane detection pipelines in C++ and MATLAB using Canny edge detection and the Hough transform.

Technical Skills

Programming: Python, C++17, C, MATLAB

Robotics: ROS, ROS2, NVIDIA Isaac Sim, NVIDIA Isaac Lab, PhysX, Newton, Gazebo, 3D Kinematics, SLAM, Reinforcement Learning

Perception: RGB-D Perception, LiDAR Point Cloud Processing, Sensor Fusion, ROS Bags, TF, OptiTrack, Computer Vision

Libraries: PCL, OpenCV, PyTorch, NumPy, relcpp, tf2, Eigen, GTSAM, Google Test

Tools: Linux, Git, Docker